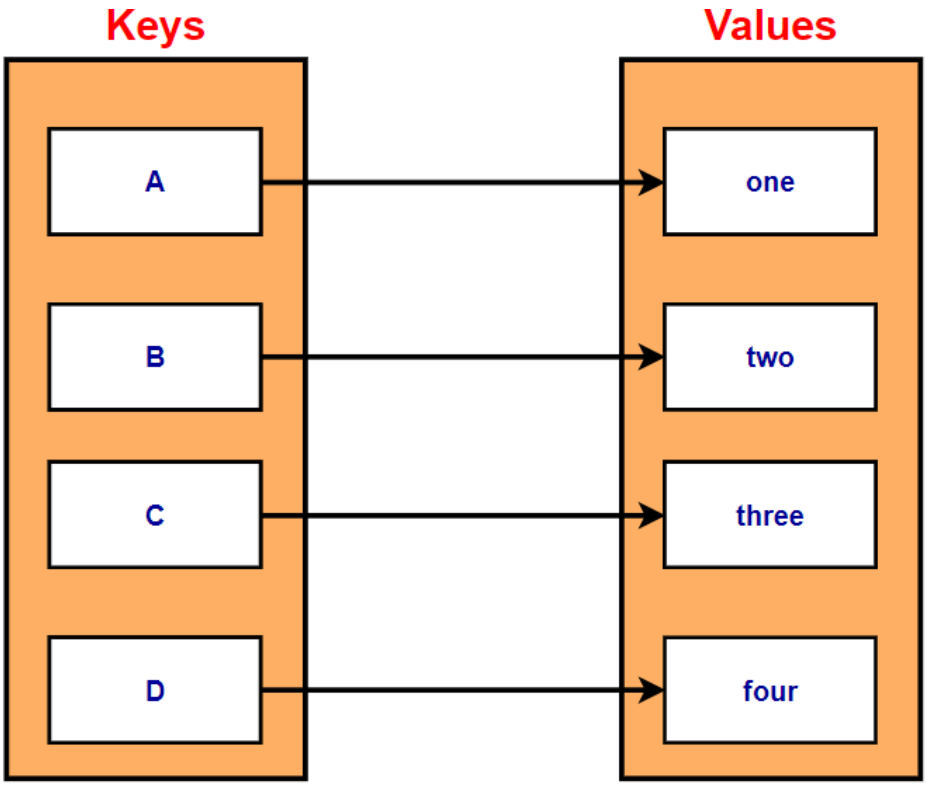
Dictionary

Dictionary is a mutable data type in Python.

A python dictionary is a collection of key and value pairs separated by a colon (:) & enclosed in curly braces {}.

Keys must be unique in a dictionary, duplicate values are allowed



mydict = {'A':'one' , 'B':'two' , 'C':'three' , 'D' :'four'}

Create Dictionary

empty dictionary

In [2]:
```
mydict = dict()
mydict
```

Out[2]: {}

In [3]:
```
mydict = {}
mydict
```

Out[3]:
```
{}
```

dictionary with integer keys

In [4]:
```
mydict = {1:'one' , 2:'two' , 3:'three'}
mydict
```

Out[4]:
```
{1: 'one', 2: 'two', 3: 'three'}
```

Create dictionary using dict()

In [6]:
```
mydict = dict({1:'one' , 2:'two' , 3:'three'})
mydict
```

Out[6]:
```
{1: 'one', 2: 'two', 3: 'three'}
```

dictionary with character keys

In [7]:
```
mydict = {'A':'one' , 'B':'two' , 'C':'three'}
mydict
```

Out[7]:
```
{'A': 'one', 'B': 'two', 'C': 'three'}
```

dictionary with mixed keys

In [8]:
```
mydict = {1:'one' , 'A':'two' , 3:'three'}
mydict
```

Out[8]:
```
{1: 'one', 'A': 'two', 3: 'three'}
```

Return Dictionary Keys using keys() method

In [9]:
```
mydict.keys()
```

Out[9]:
```
dict_keys([1, 'A', 3])
```

Return Dictionary Values using values() method

In [10]:
```
mydict.values()
```

Out[10]:
```
dict_values(['one', 'two', 'three'])
```

Access each key-value pair within a dictionary

In [11]:
```
mydict.items()
```

Out[11]:
```
dict_items([(1, 'one'), ('A', 'two'), (3, 'three')])
```

In [12]:
```
mydict = {1:'one' , 2:'two' , 'A':['asif' , 'john' , 'Maria']}
mydict
```

Out[12]:
```
{1: 'one', 2: 'two', 'A': ['asif', 'john', 'Maria']}
```

In [13]:
```
mydict = {1:'one' , 2:'two' , 'A':['asif' , 'john' , 'Maria'], 'B':('Bat','cat','ha
mydict
```

Out[13]:
```
{1: 'one',
 2: 'two',
 'A': ['asif', 'john', 'Maria'],
 'B': ('Bat', 'cat', 'hat')}
```

In [14]:
```
mydict = {1:'one' , 2:'two' , 'A':{'Name':'asif' , 'Age' :20}, 'B':('Bat','cat','ha
mydict
```

Out[14]:
```
{1: 'one',
 2: 'two',
 'A': {'Name': 'asif', 'Age': 20},
 'B': ('Bat', 'cat', 'hat')}
```

Create a dictionary from a sequence of keys

In [15]:
```
keys = {'a' , 'b' , 'c' , 'd'}
mydict3 = dict.fromkeys(keys)
mydict3
```

Out[15]:
```
{'a': None, 'c': None, 'b': None, 'd': None}
```

In [17]:
```
keys = {'a' , 'b' , 'c' , 'd'}
value = 10
mydict3 = dict.fromkeys(keys , value)
mydict3
```

Out[17]:
```
{'a': 10, 'c': 10, 'b': 10, 'd': 10}
```

In [18]:
```
keys = {'a' , 'b' , 'c' , 'd'}
value = [10,20,30]
mydict3 = dict.fromkeys(keys , value)
mydict3
```

Out[18]:
```
{'a': [10, 20, 30], 'c': [10, 20, 30], 'b': [10, 20, 30], 'd': [10, 20, 30]}
```

In [19]:
```
value.append(40)
mydict3
```

Out[19]:
```
{'a': [10, 20, 30, 40],
 'c': [10, 20, 30, 40],
 'b': [10, 20, 30, 40],
 'd': [10, 20, 30, 40]}
```

Accessing Items

In [20]:
```
mydict = {1:'one' , 2:'two' , 3:'three' , 4:'four'}
mydict
```

Out[20]:
```
{1: 'one', 2: 'two', 3: 'three', 4: 'four'}
```

Access item using key

In [21]:
```
mydict[1]
```

Out[21]:
```
'one'
```

Access item using get() method

In [22]:
```
mydict.get(1)
```

Out[22]:
```
'one'
```

In [23]:
```
mydict1 = {'Name':'Asif' , 'ID': 74123 , 'DOB': 1991 , 'job' :'Analyst'}
mydict1
```

Out[23]:
```
{'Name': 'Asif', 'ID': 74123, 'DOB': 1991, 'job': 'Analyst'}
```

In [24]:
```
mydict1['Name']
```

Out[24]:
```
'Asif'
```

In [25]:
```
mydict1.get('job')
```

Out[25]:
```
'Analyst'
```

Add, Remove & Change Items

In [26]:
```
mydict1 = {'Name':'Asif' , 'ID': 12345 , 'DOB': 1991 , 'Address' : 'Hilsinki'}
mydict1
```

Out[26]:
```
{'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Address': 'Hilsinki'}
```

Changing Dictionary Items

In [27]:
```
mydict1['DOB'] = 1992
mydict1['Address'] = 'Delhi'
mydict1
```

Out[27]:
```
{'Name': 'Asif', 'ID': 12345, 'DOB': 1992, 'Address': 'Delhi'}
```

In [28]:
```
dict1 = {'DOB':1995}
mydict1.update(dict1)
mydict1
```

Out[28]: {'Name': 'Asif', 'ID': 12345, 'DOB': 1995, 'Address': 'Delhi'}

Adding items in the dictionary

In [29]:
```
mydict1['Job'] = 'Analyst'
mydict1
```

Out[29]:
```
{'Name': 'Asif',
 'ID': 12345,
 'DOB': 1995,
 'Address': 'Delhi',
 'Job': 'Analyst'}
```

Removing items in the dictionary using Pop method

In [30]:
```
mydict1.pop('Job')
mydict1
```

Out[30]: {'Name': 'Asif', 'ID': 12345, 'DOB': 1995, 'Address': 'Delhi'}

removing a random item

In [31]:
```
mydict1.popitem()
```

Out[31]: ('Address', 'Delhi')

In [32]:
```
mydict1
```

Out[32]: {'Name': 'Asif', 'ID': 12345, 'DOB': 1995}

Removing item using del method

In [33]:
```
del[mydict1['ID']]
mydict1
```

Out[33]: {'Name': 'Asif', 'DOB': 1995}

Delete all items of the dictionary using clear method

In [34]:
```
mydict1.clear()
mydict1
```

Out[34]: {}

Delete the dictionary object

In [35]:
```
del mydict1
mydict1
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[35], line 2
      1 del mydict1
----> 2 mydict1

NameError: name 'mydict1' is not defined
```

Copy Dictionary

In [36]:
```
mydict = {'Name':'Asif' , 'ID': 12345 , 'DOB': 1991 , 'Address' : 'Hilsinki'}
mydict
```

Out[36]:
```
{'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Address': 'Hilsinki'}
```

Create a new reference "mydict1"

In [39]:
```
mydict1 = mydict
```

In [41]:
```
id(mydict) , id(mydict1) # The address of both mydict & mydict1 will be the same
```

Out[41]:
```
(2133511531520, 2133511531520)
```

Create a copy of the dictionary

In [42]:
```
mydict2 = mydict.copy()
```

In [43]:
```
id(mydict2)
```

Out[43]:
```
2133504136256
```

In [44]:
```
mydict['Address'] = 'Mumbai'
```

In [45]:
```
mydict
```

Out[45]:
```
{'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Address': 'Mumbai'}
```

In [46]:
```
mydict1 # mydict1 will be also impacted as it is pointing to the same dictionary
```

Out[46]:
```
{'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Address': 'Mumbai'}
```

In [47]:
```
mydict2 # Copy of list won't be impacted due to the changes made in the original
```

Out[47]:
```
{'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Address': 'Hilsinki'}
```

Loop through a Dictionary

In [49]:
```
mydict1 = {'Name':'Asif' , 'ID': 12345 , 'DOB': 1991 , 'Address' : 'Hilsinki' , }
mydict1
```

Out[49]:
```
{'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Address': 'Hilsinki'}
```

Key & value pair

In [51]:
```
for i in mydict1:
    print(i , ':' , mydict1[i])
```

```
Name : Asif
ID : 12345
DOB : 1991
Address : Hilsinki
```

Dictionary items

In [53]:
```
for i in mydict1:
    print(mydict1[i])
```

```
Asif
12345
1991
Hilsinki
```

Dictionary Membership

In [54]:
```
mydict1 = {'Name':'Asif' , 'ID': 12345 , 'DOB': 1991 , 'Job': 'Analyst'}
mydict1
```

Out[54]:
```
{'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Job': 'Analyst'}
```

Test if a key is in a dictionary or not.

In [55]:
```
'Name' in mydict1
```

Out[55]:
```
True
```

Membership test can be only done for keys.

In [56]:
```
'Asif' in mydict1
```

Out[56]:
```
False
```

In [57]:
```
'ID' in mydict1
```

Out[57]:
```
True
```

In [58]:
```
'Address' in mydict1
```

Out[58]:
```
False
```

All / Any

In [59]:
```
mydict1 = {'Name':'Asif' , 'ID': 12345 , 'DOB': 1991 , 'Job': 'Analyst'}
mydict1
```

Out[59]:
```
{'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Job': 'Analyst'}
```

Return false as one value is false (Value 0)

In [60]:
```
all(mydict1)
```

Out[60]:
```
True
```

In [61]:
```
s = {10, 20, 30}
```

In [3]:
```
d={1:'one', 2:'two',3:'three'}
```

In [6]:
```
d
```

Out[6]:
```
{1: 'one', 2: 'two', 3: 'three'}
```

In [7]:
```
d.keys()
```

Out[7]:
```
dict_keys([1, 2, 3])
```

In [8]:
```
d.values()
```

Out[8]:
```
dict_values(['one', 'two', 'three'])
```

In [9]:
```
d.items()
```

Out[9]:
```
dict_items([(1, 'one'), (2, 'two'), (3, 'three')])
```

In [10]:
```
d[4]='four'
d
```

Out[10]:
```
{1: 'one', 2: 'two', 3: 'three', 4: 'four'}
```

In [12]:
```
d['five']=5
d
```

Out[12]:
```
{1: 'one', 2: 'two', 3: 'three', 4: 'four', 'five': 5}
```

In [13]:
```
d['five']
```

Out[13]:
```
5
```

In [14]:
```
d[1]
```

Out[14]: 'one'

In [15]:
```
d[2]
```

Out[15]: 'two'

In [17]:
```
d.pop()
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[17], line 1
----> 1 d.pop()

TypeError: pop expected at least 1 argument, got 0
```

In [18]:
```
d.pop(2)
```

Out[18]: 'two'

In [19]:
```
d
```

Out[19]: {1: 'one', 3: 'three', 4: 'four', 'five': 5}